

21. Let the line $\frac{x}{1} = \frac{6-y}{2} = \frac{z+8}{5}$ intersect the lines $\frac{x-5}{4} = \frac{y-7}{3} = \frac{z+2}{1}$ and $\frac{x+3}{6} = \frac{3-y}{3} = \frac{z-6}{1}$ at the points A and B respectively. Then the distance of the mid-point of the line segment AB from the plane $2x - 2y + z = 14$ is :

माना रेखा $\frac{x}{1} = \frac{6-y}{2} = \frac{z+8}{5}$ रेखाओं $\frac{x-5}{4} = \frac{y-7}{3} = \frac{z+2}{1}$ तथा $\frac{x+3}{6} = \frac{3-y}{3} = \frac{z-6}{1}$ को क्रमशः बिंदुओं A तथा B पर काटती है। तो रेखाखंड AB के मध्य बिंदु की समतल $2x - 2y + z = 14$ से दूरी है :

- (A) $\frac{10}{3}$ (B) $\frac{11}{3}$ (C) 4 (D) 3

Ans. C

Sol. $\frac{x}{1} = \frac{y-6}{-2} = \frac{z+8}{5} = \lambda$ (1)

$\frac{x-5}{4} = \frac{y-7}{3} = \frac{z+2}{1} = \mu$ (2)

$\frac{x+3}{6} = \frac{y-3}{-3} = \frac{z-6}{1} = \gamma$ (3)

Intersection of (1) & (2) "A"

$(\lambda, -2\lambda + 6, 5\lambda - 8) \& (4\mu + 5, 3\mu + 7, \mu - 2)$

$\lambda = 1, \mu = -1$

A(1, 4, -3)

Intersection (1) & (3) "B"

$(\lambda, -2\lambda + 6, 5\lambda - 8) \& (6\gamma - 3, -3\gamma + 3, \gamma + 6)$

$\lambda = 1, \mu = -1$

A(1, 4, -3)

Intersection (1) & (3) "B"

$(\lambda, -2\lambda + 6, 5\lambda - 8) \& (6\gamma - 3, -3\gamma + 3, \gamma + 6)$

$\lambda = 3$

$\gamma = 1$

B(3, 0, 7)

Mid point of A & B $\Rightarrow (2, 2, 2)$

Perpendicular distance from the plane

$2x - 2y + z = 14$

$$\left| \frac{2(2) - 2(2) + 2 - 14}{\sqrt{4 + 4 + 1}} \right| = 4$$

22. For any vector $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$, with $10|a_i| < 1$, $i = 1, 2, 3$, consider the following statements :

(A) : $\max\{|a_1|, |a_2|, |a_3|\} \leq |\vec{a}|$

(B) : $|\vec{a}| \leq 3 \max\{|a_1|, |a_2|, |a_3|\}$

(A) Both (A) and (B) are true

(B) Only (B) is true

(C) Only (A) is true

(D) Neither (A) nor (B) is true

किसी सदिश $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$, $10|a_i| < 1$, $i = 1, 2, 3$ के लिए, निम्नलिखित कथनों पर विचार कीजिए :

(A) : $\max\{|a_1|, |a_2|, |a_3|\} \leq |\vec{a}|$

(B) : $|\vec{a}| \leq 3 \max\{|a_1|, |a_2|, |a_3|\}$

(A) (A) तथा (B) दोनों सत्य हैं

(B) केवल (B) सत्य है

(C) केवल (A) सत्य है

(D) न (A) और न (B) सत्य है

Ans. A

Sol. $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$, $|a_i| < \frac{1}{10}$, $i = 1, 2, 3$

$$\therefore |\vec{a}| = \sqrt{a_1^2 + a_2^2 + a_3^2} \Rightarrow |\vec{a}| = \sqrt{|a_1|^2 + |a_2|^2 + |a_3|^2} \geq \max(|a_1|, |a_2|, |a_3|)$$

$$\text{Similarly } |\vec{a}| = \sqrt{a_1^2 + a_2^2 + a_3^2} < \sqrt{\frac{1}{100} + \frac{1}{100} + \frac{1}{100}}$$

$$|\vec{a}| < \frac{\sqrt{3}}{10} < \frac{3}{10}$$

$$\Rightarrow |\vec{a}| < 3 \max(|a_1|, |a_2|, |a_3|) \Rightarrow \text{A and B both are correct}$$

23. For $a, b \in \mathbb{Z}$ and $|a - b| \leq 10$, let the angle between the plane $P: ax + y - z = b$ and the line

$l: x - 1 = a - y = z + 1$ be $\cos^{-1}\left(\frac{1}{3}\right)$. If the distance of the point $(6, -6, 4)$ from the plane P is $3\sqrt{6}$, then $a^4 + b^2$ is equal to :

$a, b \in \mathbb{Z}$ तथा $|a - b| \leq 10$ के लिए, माना समतल $P: ax + y - z = b$ तथा रेखा $l: x - 1 = a - y = z + 1$ के बीच का कोण

$\cos^{-1}\left(\frac{1}{3}\right)$ है। यदि बिंदु $(6, -6, 4)$ की समतल P से दूरी $3\sqrt{6}$ है, तो $a^4 + b^2$ बराबर है :

(A) 25

(B) 85

(C) 48

(D) 32

Ans. D

Sol. $ax + y - z = b$

$$\frac{x-1}{1} = \frac{y-a}{-1} = \frac{z+1}{1}$$

$$\vec{n} = a\hat{i} + \hat{j} - \hat{k}$$

$$\vec{p} = \hat{i} - \hat{j} + \hat{k}$$

$$\sin \theta = \frac{|\vec{n} \cdot \vec{p}|}{|\vec{n}| |\vec{p}|}$$

$$\left| \frac{a-1-1}{\sqrt{a^2+2}\sqrt{3}} \right| = \frac{2\sqrt{2}}{3}$$

$$3(a-2)^2 = 8 \times 3(a^2+2)$$

$$5a^2 + 12a + 4 = 0 \Rightarrow (5a+2)(a+2) = 0$$

$$a = -2 \quad \& \quad a = -2/5 \text{ (reject)}$$

$$a = -2, \quad a - b \leq 10$$

$$b \geq -8n$$

Now distance of given point from the plane

$$-2x + y - z = b$$

$$\frac{12+6+4+b}{\sqrt{6}} = \pm 3\sqrt{6}$$

$$b+22 = \pm 18$$

$$b = -4 \text{ and } b = -40 \text{ (Reject)}$$

$$a^4 + b^2 = 16 + 16 = 32$$

24. Let the position vectors of two points P and Q be $3\hat{i} - \hat{j} + 2\hat{k}$ and $\hat{i} + 2\hat{j} - 4\hat{k}$, respectively. Let R and S be two points such that the direction ratios of lines PR and QS are $(4, -1, 2)$ and $(-2, 1, -2)$, respectively. Let lines PR and QS intersect at T. If the vector $\overrightarrow{TA}$ is perpendicular to both $\overrightarrow{PR}$ and $\overrightarrow{QS}$ and the length of vector $\overrightarrow{TA}$ is $\sqrt{5}$ units, then modulus of a position vector of A is :

माना दो बिन्दुओं P तथा Q के स्थिति सदिश क्रमशः $3\hat{i} - \hat{j} + 2\hat{k}$ तथा $\hat{i} + 2\hat{j} - 4\hat{k}$ है। माना दो बिन्दु R तथा S इस प्रकार हैं कि रेखाओं PR तथा QS के दिक् अनुपात क्रमशः $(4, -1, 2)$ तथा $(-2, 1, -2)$ हैं। माना रेखाओं PR तथा QS का प्रतिच्छेदन बिन्दु T है। यदि सदिश $\overrightarrow{TA}$, सदिशों $\overrightarrow{PR}$ तथा $\overrightarrow{QS}$ के लम्बवत् है तथा सदिश $\overrightarrow{TA}$ की लम्बाई $\sqrt{5}$ इकाई है, तो A के एक सदिश का मापांक है :

- (A) $\sqrt{171}$ (B) $\sqrt{227}$ (C) $\sqrt{482}$ (D) $\sqrt{5}$

Ans. A

Sol. Equation of line

PR $\vec{r} = (3, -1, 2) + \lambda (4, -1, 2)$

QS : $\vec{r} = (1, 2, -4) + \mu (-2, 1, -2)$

POI $\equiv T$

$3 + 4\lambda = 1 - 2\mu$

$-1 - \lambda = 2 + \mu \Rightarrow \lambda = 2, \mu = -5$

$2 + 2\lambda = -4 - 2\mu$

$T(11, -3, 6)$

$\vec{n}$ = vector parallel to $\overrightarrow{TA}$

$$= \overrightarrow{PR} \times \overrightarrow{QS} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 4 & -1 & 2 \\ -2 & 1 & -2 \end{vmatrix}$$

$= 4\hat{j} + 2\hat{k}$

equation of $\overrightarrow{TA}$: $\vec{r} = (11, -3, 6) + \lambda (0, 2, 1)$

$|\overrightarrow{TA}| = \sqrt{5} \Rightarrow \lambda = \pm 1$

A $(11, -1, 7)$ or A $(11, -5, 5)$

$|\overrightarrow{OA}| = \sqrt{121+1+49} = \sqrt{171}$

25. If a point $P(\alpha, \beta, \gamma)$ satisfying $(\alpha \ \beta \ \gamma) \begin{pmatrix} 2 & 10 & 8 \\ 9 & 3 & 8 \\ 8 & 4 & 8 \end{pmatrix} = (0 \ 0 \ 0)$ lies on the plane $2x + 4y + 3z = 5$, then

$6\alpha + 9\beta + 7\gamma$ is equal to :

यदि $(\alpha \ \beta \ \gamma) \begin{pmatrix} 2 & 10 & 8 \\ 9 & 3 & 8 \\ 8 & 4 & 8 \end{pmatrix} = (0 \ 0 \ 0)$ को संतुष्ट करने वाला एक बिंदु $P(\alpha, \beta, \gamma)$ समतल $2x + 4y + 3z = 5$ पर है, तो

$6\alpha + 9\beta + 7\gamma$ बराबर है :

- (A) -1 (B) 11 (C) $\frac{11}{5}$ (D) $\frac{5}{4}$

Ans. B

Sol. $(\alpha \ \beta \ \gamma) \begin{pmatrix} 2 & 10 & 8 \\ 9 & 3 & 8 \\ 8 & 4 & 8 \end{pmatrix} = (0 \ 0 \ 0)$

$$2\alpha + 9\beta + 8\gamma = 0 \quad \text{_____} (1)$$

$$10\alpha + 3\beta + 4\gamma = 0 \quad \text{_____} (2)$$

$$8\alpha + 8\beta + 8\gamma = 0 \quad \text{_____} (3)$$

from (1) and (3) by cross multiplication

$$\frac{\alpha}{1} = \frac{\beta}{6} = \frac{\gamma}{-7} = k \Rightarrow \alpha = k, \beta = 6k, \gamma = -7k$$

$$\begin{aligned} \text{Point } P(\alpha, \beta, \gamma) \text{ lie on the plane } 2x + 4y + 3z = 5 &\Rightarrow 2\alpha + 4\beta + 3\gamma = 5 \\ &\Rightarrow 2k + 24k - 21k = 5 \\ &\Rightarrow k = 1 \end{aligned}$$

$$\text{Hence } 6\alpha + 9\beta + 7\gamma = 6k + 54k - 49k = 11k = 11$$

26. The foot of perpendicular from the origin O to a plane P which meets the co-ordinate axes at the points A, B, C is $(2, a, 4)$, $a \in \mathbb{N}$. If the volume of the tetrahedron OABC is 144, then which of the following points is **NOT** on P?

मूल बिंदु से एक समतल P, जो निर्देशांक अक्षों को बिंदुओं A, B, C पर मिलता है, पर डाले गये लंब का पाद $(2, a, 4)$, $a \in \mathbb{N}$ है। यदि चतुष्फलक OABC का आयतन 144 है, तो निम्न बिंदुओं में से कौन सा P पर नहीं है ?

- (A) (0,4,4) (B) (2,2,4) (C) (0,6,3) (D) (3,0,4)

Ans. D

Sol. Equation of Plane :

$$(2\hat{i} + a\hat{j} + 4\hat{k}) \cdot [(x-2)\hat{i} + (y-a)\hat{j} + (z-4)\hat{k}] = 0$$

$$\Rightarrow 2x + ay + 4z = 20 + a^2$$

$$\Rightarrow A \equiv \left(\frac{20+a^2}{2}, 0, 0 \right)$$

$$B \equiv \left(0, \frac{20+a^2}{a}, 0 \right)$$

$$C \equiv \left(0, 0, \frac{20+a^2}{4} \right)$$

$\Rightarrow$ Volume of tetrahedron

$$= \frac{1}{6} [\vec{a} \cdot \vec{bc}]$$

$$= \frac{1}{6} \vec{a} (\vec{b} \times \vec{c})$$

$$\Rightarrow \frac{1}{6} \left(\frac{20+a^2}{2} \right) \cdot \left(\frac{20+a^2}{a} \right) \cdot \left(\frac{20+a^2}{4} \right) = 144$$

$$\Rightarrow (20+a^2)^3 = 144 \times 48 \times a$$

$$\Rightarrow a = 2$$

$$\Rightarrow \text{Equation of plane is } 2x + 2y + 4z = 24$$

$$\text{Or } x + y + 2z = 12$$

$$\Rightarrow (3, 0, 4) \text{ Not lies on the Plane } x + y + 2z = 12$$

27. Let $\sqrt{3}\hat{i} + \hat{j}, \hat{i} + \sqrt{3}\hat{j}$ and $\beta\hat{i} + (1-\beta)\hat{j}$ respectively be the position vectors of the points A, B and C with respect to the origin O. If the distance of C from the bisector of the acute angle between OA and OB is $\frac{3}{\sqrt{2}}$, then the sum of all possible values of β is :

माना $\sqrt{3}\hat{i} + \hat{j}, \hat{i} + \sqrt{3}\hat{j}$ तथा $\beta\hat{i} + (1-\beta)\hat{j}$ क्रमशः तीन बिन्दुओं A, B तथा C के मूलबिन्दु O के सापेक्ष, स्थिति सदिश हैं। यदि C

की, OA तथा OB के बीच बने न्यूनकोण के समद्विभाजक से दूरी $\frac{3}{\sqrt{2}}$ है, तो β के सभी संभावित मानों का योग है -

(A) 2

(B) 1

(C) 3

(D) 4

Ans. B

Sol. Angle bisector is $x - y = 0$

$$\left| \frac{\beta - (1 - \beta)}{\sqrt{2}} \right| = \frac{3}{\sqrt{2}}$$

$$|2\beta - 1| = 3$$

$$2\beta - 1 = 3 \text{ or } 2\beta - 1 = -3$$

$$2\beta = 4 \quad 2\beta = -2$$

$$\beta = 2 \quad \beta = -1$$

28. Let $\vec{a} = 2\hat{i} + \hat{j} + \hat{k}$, and $\vec{b}$ and $\vec{c}$ be two non zero vectors such that $|\vec{a} + \vec{b} + \vec{c}| = |\vec{a} + \vec{b} - \vec{c}|$ and $\vec{b} \cdot \vec{c} = 0$. Consider the following two statements :

(A) $|\vec{a} + \lambda\vec{c}| \geq |\vec{a}|$ for all $\lambda \in \mathbb{R}$.

(B) $\vec{a}$ and $\vec{c}$ are always parallel.

Then,

(A) both (A) and (B) are correct

(B) only (A) is correct

(C) only (B) is correct

(D) neither (A) nor (B) is correct.

माना $\vec{a} = 2\hat{i} + \hat{j} + \hat{k}$ तथा $\vec{b}$ और $\vec{c}$ दो शून्येतर सदिश हैं जिनके लिए $|\vec{a} + \vec{b} + \vec{c}| = |\vec{a} + \vec{b} - \vec{c}|$ तथा $\vec{b} \cdot \vec{c} = 0$ है। निम्न दो कथनों का विचार कीजिए :

(A) $\lambda \in \mathbb{R}$ के लिए $|\vec{a} + \lambda\vec{c}| \geq |\vec{a}|$

(B) $\vec{a}$ तथा $\vec{c}$ सदा समांतर है

तो,

(A) (A) तथा (B) सही है

(B) केवल (A) सही है

(C) केवल (B) सही है

(D) न तो (A) न ही (B) सही है

Ans. B

Sol. $|\vec{a} + \vec{b} + \vec{c}|^2 = |\vec{a} + \vec{b} - \vec{c}|^2$

$$2\vec{a} \cdot \vec{b} + 2\vec{b} \cdot \vec{c} + 2\vec{c} \cdot \vec{a} = 2\vec{a} \cdot \vec{b} - 2\vec{b} \cdot \vec{c} - 2\vec{c} \cdot \vec{a}$$

$$4\vec{a} \cdot \vec{c} = 0$$

B is incorrect.

$$|\vec{a} + \lambda\vec{c}|^2 \geq |\vec{a}|^2$$

$$\lambda^2 c^2 \geq 0$$

True $\forall \lambda \in \mathbb{R}$ (A) is correct.

29. Let the vectors $\vec{u}_1 = \hat{i} + \hat{j} + a\hat{k}$, $\vec{u}_2 = \hat{i} + b\hat{j} + \hat{k}$ and $\vec{u}_3 = c\hat{i} + \hat{j} + \hat{k}$ be coplanar. If the vectors $\vec{v}_1 = (a+b)\hat{i} + c\hat{j} + c\hat{k}$, $\vec{v}_2 = a\hat{i} + (b+c)\hat{j} + a\hat{k}$ and $\vec{v}_3 = b\hat{i} + b\hat{j} + (c+a)\hat{k}$ are also coplanar, then $6(a+b+c)$ is equal to :

माना सदिश $\vec{u}_1 = \hat{i} + \hat{j} + a\hat{k}$, $\vec{u}_2 = \hat{i} + b\hat{j} + \hat{k}$ तथा $\vec{u}_3 = c\hat{i} + \hat{j} + \hat{k}$ सह-तलीय है। यदि सदिश $\vec{v}_1 = (a+b)\hat{i} + c\hat{j} + c\hat{k}$, $\vec{v}_2 = a\hat{i} + (b+c)\hat{j} + a\hat{k}$ तथा $\vec{v}_3 = b\hat{i} + b\hat{j} + (c+a)\hat{k}$ भी सह-तलीय हैं, तो $6(a+b+c)$ बराबर है :

- (A) 6 (B) 4 (C) 12 (D) 0

Ans. C

Sol. Given: $\vec{u}_1 = \hat{i} + \hat{j} + a\hat{k}$, $\vec{u}_2 = \hat{i} + b\hat{j} + \hat{k}$ and $\vec{u}_3 = c\hat{i} + \hat{j} + \hat{k}$

and $\vec{v}_1 = (a+b)\hat{i} + c\hat{j} + c\hat{k}$, $\vec{v}_2 = a\hat{i} + (b+c)\hat{j} + a\hat{k}$ and $\vec{v}_3 = b\hat{i} + b\hat{j} + (c+a)\hat{k}$

$$\text{Now } \begin{vmatrix} 1 & 1 & a \\ 1 & b & 1 \\ c & 1 & 1 \end{vmatrix} = 0 \Rightarrow (b-1) - (1-c) + a(1-bc) = 0$$

$$a + b + c = 2 + abc \quad \dots(i)$$

$$\text{and } \begin{vmatrix} a+b & c & c \\ a & b+c & a \\ b & b & c+a \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} a+b & c & c \\ a & b+c & a \\ -2a & -2c & 0 \end{vmatrix} = 0 \quad (R_3 \rightarrow R_3 - (R_1 + R_2))$$

$$\Rightarrow abc = 0$$

$$\therefore a + b + c = 2$$

$$\therefore 6(a+b+c) = 12$$

30. Let $\vec{a}$ be a non-zero vector parallel to the line of intersection of the two planes described by $\hat{i} + \hat{j}, \hat{i} + \hat{k}$ and $\hat{i} - \hat{j}, \hat{j} - \hat{k}$. If θ is the angle between the vector $\vec{a}$ and the vector $\vec{b} = 2\vec{i} - 2\vec{j} + \vec{k}$ and $\vec{a} \cdot \vec{b} = 6$, then the ordered pair $(\theta, |\vec{a} \times \vec{b}|)$ is equal to :

माना $\vec{a}$; $\hat{i} + \hat{j}, \hat{i} + \hat{k}$ तथा $\hat{i} - \hat{j}, \hat{j} - \hat{k}$ द्वारा वर्णित दो समतलों के दो प्रतिच्छेदन रेखा के समान्तर एक अशून्य सदिश है। यदि सदिश $\vec{a}$ तथा सदिश $\vec{b} = 2\vec{i} - 2\vec{j} + \vec{k}$ के बीच का कोण θ है तो क्रमित युग्म $(\theta, |\vec{a} \times \vec{b}|)$ का मान ज्ञात कीजिए :

- (A) $\left(\frac{\pi}{4}, 6\right)$ (B) $\left(\frac{\pi}{3}, 6\right)$ (C) $\left(\frac{\pi}{4}, 3\sqrt{6}\right)$ (D) $\left(\frac{\pi}{3}, 3\sqrt{6}\right)$

Ans. A

Sol. $\vec{n}_1$ and $\vec{n}_2$ are normal vector to the plane $\hat{i} + \hat{j}, \hat{i} + \hat{k}$ and $\hat{i} - \hat{j}, \hat{i} - \hat{k}$ respectively

$$\vec{n}_1 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{vmatrix} = \hat{i} - \hat{j} - \hat{k}$$

$$\vec{n}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -1 & 0 \\ 1 & 0 & -1 \end{vmatrix} = \hat{i} + \hat{j} + \hat{k}$$

$$\vec{a} = \lambda |\vec{n}_2 \times \vec{n}_1|$$

$$= \lambda \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -1 & -1 \\ 1 & 1 & 1 \end{vmatrix} = \lambda (-2\hat{j} + 2\hat{k})$$

$$\vec{a} \cdot \vec{b} = \lambda |0 + 4 + 2| = 6$$

$$\Rightarrow \lambda = 1$$

$$\vec{a} = -2\hat{j} + 2\hat{k}$$

$$\cos \theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|}$$

$$\cos \theta = \frac{6}{2\sqrt{2} \times 3} = \frac{1}{\sqrt{2}}$$

$$\theta = \frac{\pi}{4}$$

$$\text{Now } |\vec{a} \cdot \vec{b}|^2 + |\vec{a} \times \vec{b}|^2 = |\vec{a}|^2 |\vec{b}|^2$$

$$36 + |\vec{a} \times \vec{b}|^2 = 8 \times 9 = 72$$

$$|\vec{a} \times \vec{b}|^2 = 36$$

$$|\vec{a} \times \vec{b}| = 6$$

31. Let $\vec{a} = 3\hat{i} + 2\hat{j} + x\hat{k}$ and $\vec{b} = \hat{i} - \hat{j} + \hat{k}$, for some real x . Then $|\vec{a} \times \vec{b}| = r$ is possible if :

माना किसी वास्तविक संख्या x के लिए $\vec{a} = 3\hat{i} + 2\hat{j} + x\hat{k}$ तथा $\vec{b} = \hat{i} - \hat{j} + \hat{k}$ है, तो $|\vec{a} \times \vec{b}| = r$ तभी सम्भव है, जब :

(A) $3\sqrt{\frac{3}{2}} < r < 5\sqrt{\frac{3}{2}}$ (B) $\sqrt{\frac{3}{2}} < r \leq 3\sqrt{\frac{3}{2}}$ (C) $0 < r \leq \sqrt{\frac{3}{2}}$ (D) $r \geq 5\sqrt{\frac{3}{2}}$

Ans. D

Sol. $\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 2 & x \\ 1 & -1 & 1 \end{vmatrix} = (2+x)\hat{i} + (x-3)\hat{j} - 5\hat{k}$

$$|\vec{a} \times \vec{b}| = r = \sqrt{(2+x)^2 + (x-3)^2 + (-5)^2}$$

$$\Rightarrow r = \sqrt{4 + x^2 + 4x + x^2 + 9 - 6x + 25}$$

$$= \sqrt{2x^2 - 2x + 38} = \sqrt{2\left(x^2 - x + \frac{1}{4}\right) + 38 - \frac{1}{2}}$$

$$= \sqrt{2\left(x - \frac{1}{2}\right)^2 + \frac{75}{2}}$$

$$\Rightarrow r \geq \sqrt{\frac{75}{2}} \Rightarrow r \geq 5\sqrt{\frac{3}{2}}$$

32. Let the equation of plane passing through the line of intersection of the planes $x + 2y + az = 2$ and $x - y + z = 3$ be $5x - 11y + bz = 6a - 1$. For $c \in \mathbb{Z}$, if the distance of this plane from the point $(a, -c, c)$ is $\frac{2}{\sqrt{a}}$,

then $\frac{a+b}{c}$ is equal to :

माना समतलों $x + 2y + az = 2$ तथा $x - y + z = 3$ की प्रतिच्छेदन रेखा से होकर जाने वाले समतल का समीकरण

$5x - 11y + bz = 6a - 1$ है। $c \in \mathbb{Z}$ के लिए, यदि इस समतल की बिन्दु $(a, -c, c)$ से दूरी $\frac{2}{\sqrt{a}}$ है, तो $\frac{a+b}{c}$ बराबर है :

(A) -4 (B) -2 (C) 4 (D) 2

Ans. A

Sol. $(x + 2y + az - 2) + \lambda (x - y + z - 3) = 0$

$$\frac{1+\lambda}{5} = \frac{2-\lambda}{-11} = \frac{a+\lambda}{b} = \frac{2+3\lambda}{6a-1}$$

$$\lambda = -\frac{7}{2}, a = 3, b = 1$$

$$\frac{2}{\sqrt{a}} = \frac{|5a+11c+bc-6a+1|}{\sqrt{25+121+1}}$$

$$c = -1$$

$$\therefore \frac{a+b}{c} = \frac{3+1}{-1} = -4$$

33. Let a vector $\alpha\hat{i} + \beta\hat{j}$ be obtained by rotating the vector $\sqrt{3}\hat{i} + \hat{j}$ by an angle 45° about the origin in counter-clockwise direction in the first quadrant. Then the area of triangle having vertices (α, β) , $(0, \beta)$ and $(0, 0)$ is equal to:

माना मूल बिन्दु के सापेक्ष सदिश $\sqrt{3}\hat{i} + \hat{j}$ को प्रथम चतुर्थांश में वामावर्त दिशा में 45° के कोण तक घुमाने पर सदिश $\alpha\hat{i} + \beta\hat{j}$ प्राप्त होता है। तो शीर्ष (α, β) , $(0, \beta)$ तथा $(0, 0)$ के त्रिभुज का क्षेत्रफल बराबर है :

- (A) $\frac{1}{\sqrt{2}}$ (B) 1 (C) $2\sqrt{2}$ (D) $\frac{1}{2}$

Ans. D

Sol. $\alpha\hat{i} + \beta\hat{j} = 2(\cos 75^\circ \hat{i} + \sin 75^\circ \hat{j})$

$$\text{Area} = \frac{1}{2} \begin{vmatrix} \alpha & \beta & 1 \\ 0 & \beta & 1 \\ 0 & 0 & 1 \end{vmatrix} = \left| \frac{1}{2} \alpha \beta \right|$$

$$= \frac{1}{2} |2 \cdot 2 \sin 75^\circ \cos 75^\circ| = \frac{2 \sin 150^\circ}{2} = \frac{1}{2}$$

34. One vertex of a rectangular parallelepiped is at the origin O and the lengths of its edges along x, y and z axes are 3, 4 and 5 units respectively. Let P be the vertex (3,4,5). Then the shortest distance between the diagonal OP and an edge parallel to z axis, not passing through O or P is:

एक आयताकार समांतर षट्फलक का एक शीर्ष मूल बिंदु O पर है और x, y तथा z अक्षों के अनुदिश इसके किनारों (edges) की लम्बाईयों क्रमशः 3, 4 तथा 5 इकाई है। माना इसका शीर्ष P बिंदु (3,4,5) पर है। तब विकर्ण OP तथा z अक्ष के समांतर इसके एक किनारे, जो O या P से होकर नहीं जाता है, के बीच न्यूनतम दूरी है :

(A) $\frac{12}{5\sqrt{5}}$

(B) $\frac{12}{\sqrt{5}}$

(C) $12\sqrt{5}$

(D) $\frac{12}{5}$

Ans. D

Sol. Equation of OP is $\frac{x}{3} = \frac{y}{4} = \frac{z}{5}$

$a_1 = (0, 0, 0) \quad a_2 = (3, 0, 5)$

$b_1 = (3, 4, 5) \quad b_2 = (0, 0, 1)$

Equation of edge parallel to z axis

$$\frac{x-3}{0} = \frac{y-0}{0} = \frac{z-5}{1}$$

$$S.D = \frac{(\vec{a}_2 \cdot \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2)}{|\vec{b}_1 \times \vec{b}_2|}$$

$$\frac{\begin{vmatrix} 3 & 0 & 5 \\ 3 & 4 & 5 \\ 0 & 0 & 1 \end{vmatrix}}{\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 4 & 5 \\ 0 & 0 & 1 \end{vmatrix}} = \frac{3(4)}{|4\hat{i} - 3\hat{j}|} = \frac{12}{5}$$

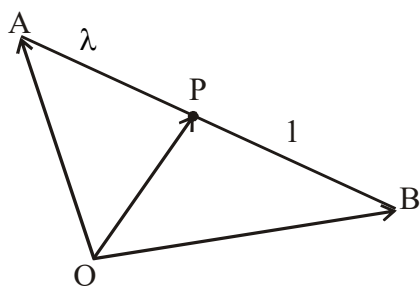
INTEGER TYPE QUESTIONS

35. Let the position vectors of points 'A' and 'B' be $\hat{i} + \hat{j} + \hat{k}$ and $2\hat{i} + \hat{j} + 3\hat{k}$, respectively. A point 'P' divides the line segment AB internally in the ratio $\lambda : 1$ ($\lambda > 0$). If O is the origin and $\vec{OB} \cdot \vec{OP} - 3|\vec{OA} \times \vec{OP}|^2 = 6$, then λ is equal to _____.

माना बिन्दुओं 'A' तथा 'B' के स्थिति सदिश क्रमशः $\hat{i} + \hat{j} + \hat{k}$ तथा $2\hat{i} + \hat{j} + 3\hat{k}$ हैं। एक बिन्दु P, रेखा खंड AB को अन्तः अनुपात $\lambda : 1$ ($\lambda > 0$) में विभाजित करता है। यदि O मूल बिन्दु है तथा $\vec{OB} \cdot \vec{OP} - 3|\vec{OA} \times \vec{OP}|^2 = 6$ है, तो λ बराबर है _____।

Ans. 0.80

Sol.



$$\overrightarrow{OP} = \frac{\lambda \overrightarrow{OB} + \overrightarrow{OA}}{\lambda + 1}$$

$$\overrightarrow{OB} \cdot \overrightarrow{OP} = \frac{\lambda |\overrightarrow{OB}|^2 + \overrightarrow{OA} \cdot \overrightarrow{OB}}{\lambda + 1}$$

$$= \frac{\lambda (\sqrt{14})^2 + (2 + 1 + 3)}{\lambda + 1} = \frac{6 + 14\lambda}{\lambda + 1}$$

$$= \frac{\lambda}{\lambda + 1} \begin{vmatrix} 1 & j & k \\ 1 & 1 & 1 \\ 2 & 1 & 3 \end{vmatrix} = \frac{\lambda}{\lambda + 1} (2\hat{i} - \hat{j} - \hat{k})$$

$$|\overrightarrow{OA} \times \overrightarrow{OP}|^2 = \frac{\lambda^2}{(\lambda + 1)^2} (6)$$

Use given equation

$$\frac{6 + 14\lambda}{\lambda + 1} - 3 \cdot \frac{6\lambda^2}{(\lambda + 1)^2} = 6$$

$$\Rightarrow \frac{8\lambda}{\lambda + 1} = 18 \left(\frac{\lambda}{\lambda + 1} \right)^2$$

$$\Rightarrow 8 = 18 \cdot \frac{\lambda}{\lambda + 1} \quad \left[\frac{\lambda}{\lambda + 1} \neq 0 \right]$$

$$\Rightarrow 8\lambda + 8 = 18\lambda$$

$$\therefore \lambda = \frac{8}{10} = 0.8$$

36. If the equation of the plane passing through the point (1,1,2) and are perpendicular to the line $x - 3y + 2z - 1 = 0 = 4x - y + z$ is $Ax + By + Cz = 1$, then $140(C - B + A)$ is equal to _____ .

यदि बिंदु $(1,1,2)$ से होकर जाने वाला तथा रेखा $x - 3y + 2z - 1 = 0 = 4x - y + z$ के लंबवत् समतल का समीकरण $Ax + By + Cz = 1$ है, तो $140(C - B + A)$ बराबर है।

Ans. 15

Sol. $x - 3y + 2z - 1 = 0$
 $4x - y + z = 0$

$$\vec{n}_1 \times \vec{n}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -3 & 2 \\ 4 & -1 & 1 \end{vmatrix}$$

$$= -\hat{i} + 7\hat{j} + 11\hat{k}$$

D.R. of normal to the plane is $-1, 7, 11$

Equation of plane

$$-1(x - 1) + 7(y - 1) + 11(z - 2) = 0$$

$$-x + 7y + 11z = 28$$

$$-\frac{1}{28}x + \frac{7}{28}y + \frac{11}{28}z = 1$$

$$Ax + By + Cz = 1$$

$$140(C - B + A) = 140\left(\frac{11}{28} - \frac{7}{28} - \frac{1}{28}\right)$$

$$= 140 \times \frac{3}{28} = 15$$

37. If the line $x = y = z$ intersects the line $x \sin A + y \sin B + z \sin C - 18 = 0 = x \sin 2A + y \sin 2B + z \sin 2C - 9$, where A, B, C are the angles of a triangle ABC , then $80\left(\sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}\right)$ is equal to _____.

यदि रेखा $x = y = z$ रेखाओं $x \sin A + y \sin B + z \sin C - 18 = 0 = x \sin 2A + y \sin 2B + z \sin 2C - 9$ को प्रतिच्छेद करती है जहाँ A, B, C त्रिभुज ABC के कोण हैं, तो $80\left(\sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}\right)$ का मान होगा _____

Ans. 5

Sol. $x = y = z = k$ (let

$$\therefore k(\sin A + \sin B + \sin C) = 18$$

$$\Rightarrow k\left(4 \cos \frac{A}{2} \cos \frac{B}{2} \cos \frac{C}{2}\right) = 18 \quad \dots(i)$$

$$k(\sin 2A + \sin 2B + \sin 2C) = 9$$

$$\Rightarrow k(4 \sin A \cdot \sin B \cdot \sin C) = 9$$

...(ii)

(ii)/(i)

$$8 \sin \frac{A}{2} \cdot \sin \frac{B}{2} \cdot \sin \frac{C}{2} = \frac{9}{18}$$

$$\Rightarrow 80 \sin \frac{A}{2} \cdot \sin \frac{B}{2} \cdot \sin \frac{C}{2} = 5$$

38. Let the foot of perpendicular from the point $A(4, 3, 1)$ on the plane $P : x - y + 2z + 3 = 0$ be N . If $B(5, \alpha, \beta)$, $\alpha, \beta \in \mathbb{Z}$ is a point on plane P such that the area of the triangle ABN is $3\sqrt{2}$, then $\alpha^2 + \beta^2 + \alpha\beta$ is equal to _____.

माना बिंदु $A(4, 3, 1)$ से समतल $P : x - y + 2z + 3 = 0$ पर डाला गया लंबपाद N है। यदि समतल P पर एक बिंदु $B(5, \alpha, \beta)$, $\alpha, \beta \in \mathbb{Z}$ इस प्रकार है कि त्रिभुज ABN का क्षेत्रफल $3\sqrt{2}$ है, तो $\alpha^2 + \beta^2 + \alpha\beta$ बराबर है :

Ans. 7

Sol. $5 - \alpha + 2\beta + 3 = 0$

$$\alpha - 2\beta = 8$$

$$\frac{x_1 - 4}{1} = \frac{y_1 - 3}{-1} = \frac{z_1 - 1}{2} = \frac{-(6)}{6}$$

$$N \equiv (x_1, y_1, z_1) = (3, 4, -1)$$

$$B \equiv (5, \alpha, \beta)$$

$$\frac{1}{2} AN \times BN = 3\sqrt{2}$$

$$AN \cdot BN = 6\sqrt{2}$$

$$\frac{6}{\sqrt{6}} \cdot BN = 6\sqrt{2}$$

$$BN = 2\sqrt{3}$$

$$4 + (\alpha - 4)^2 + (\beta + 1)^2 = 12$$

$$(\alpha - 4)^2 + (\beta + 1)^2 = 8$$

$$(2\beta + 4)^2 + (\beta + 1)^2 = 8$$

$$5\beta^2 + 18\beta + 9 = 0$$

$$\beta = -3, -\frac{3}{5}$$

$$\beta = -3, \alpha = 2$$

$$\alpha^2 + \beta^2 + \alpha\beta = 9 + 4 - 6 = 7$$

39. Let $\vec{v} = \alpha\hat{i} + 2\hat{j} - 3\hat{k}$, $\vec{w} = 2\alpha\hat{i} + \hat{j} - \hat{k}$ and $\vec{u}$ be a vector such that $|\vec{u}| = \alpha > 0$. If the minimum value of the scalar triple product $[\vec{u} \vec{v} \vec{w}]$ is $-\alpha\sqrt{3401}$, and $|\vec{u} \cdot \hat{i}|^2 = \frac{m}{n}$ where m and n are coprime natural numbers, then m + n is equal to _____.

माना $\vec{v} = \alpha\hat{i} + 2\hat{j} - 3\hat{k}$, $\vec{w} = 2\alpha\hat{i} + \hat{j} - \hat{k}$ है तथा एक सदिश $\vec{u}$ के लिए $|\vec{u}| = \alpha > 0$ है। यदि अदिश त्रिक गुणनफल $[\vec{u} \vec{v} \vec{w}]$ का न्यूनतम मान $-\alpha\sqrt{3401}$ है, और $|\vec{u} \cdot \hat{i}|^2 = \frac{m}{n}$ है जहाँ m एवं n सहअभाज्य धन पूर्णांक हैं, तब m + n का मान है।

Ans. 3501

Sol. $[\vec{u} \vec{v} \vec{w}] = \vec{u} \cdot (\vec{v} \times \vec{w})$

$$(|u||\vec{v} \times \vec{w}| \cos \theta) = -\alpha\sqrt{3401}$$

$$\Rightarrow \cos \theta = -1$$

$$|u| = \alpha \text{ (Given)}$$

$$|\vec{v} \times \vec{w}| = \sqrt{3401}$$

$$\vec{v} \times \vec{w} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \alpha & 2 & -3 \\ 2\alpha & 1 & -1 \end{vmatrix}$$

$$\vec{v} \times \vec{w} = \hat{i} - 5\alpha\hat{j} - 3\alpha\hat{k}$$

$$|\vec{v} \times \vec{w}| = \sqrt{1 + 25\alpha^2 + 9\alpha^2} = \sqrt{3401}$$

$$34\alpha^2 = 3400$$

$$\alpha^2 = 100$$

$$\alpha = 10 \quad (\text{as } \alpha > 0)$$

$$\text{So } \vec{u} = \lambda(\hat{i} - 5\alpha\hat{j} - 3\alpha\hat{k})$$

$$|\vec{u}| = \sqrt{\lambda^2 + 25\alpha^2\lambda^2 + 9\alpha^2\lambda^2}$$

$$\alpha^2 = \lambda^2(1 + 25\alpha^2 + 9\alpha^2)$$

$$100 = \lambda^2(1 + 34 \times 100)$$

$$\lambda^2 = \frac{100}{3401} = \frac{m}{n}$$

40. Let the plane $x + 3y - 2z + 6 = 0$ meet the co-ordinate axes at the points A, B, C. If the orthocenter of the triangle ABC is $\left(\alpha, \beta, \frac{6}{7}\right)$, then $98(\alpha + \beta)^2$ is equal to _____.

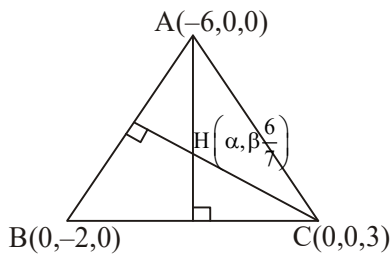
माना समतल $x + 3y - 2z + 6 = 0$ निर्देशांक अक्षों को बिंदुओं A, B, C पर मिलता है। यदि त्रिभुज ABC का लंबकेन्द्र $\left(\alpha, \beta, \frac{6}{7}\right)$ है, तो $98(\alpha + \beta)^2$ बराबर है _____

Ans. 288

Sol. A(-6, 0, 0) B(0, -2, 0) C = (0, 0, 3)

$$\overrightarrow{AB} = 6\hat{i} - 2\hat{j}, \overrightarrow{BC} = 2\hat{j} + 3\hat{k},$$

$$\overrightarrow{AC} = 6\hat{i} + 3\hat{k}$$



$$\overrightarrow{AH} \cdot \overrightarrow{BC} = 0$$

$$\left(\alpha + 6, \beta, \frac{6}{7}\right) \cdot (0, 2, 3) = 0$$

$$\beta = \frac{-9}{7}$$

$$\overrightarrow{CH} \cdot \overrightarrow{AB} = 0$$

$$\left(\alpha, \beta, \frac{-15}{7}\right) \cdot (6, -2, 0) = 0$$

$$6\alpha - 2\beta = 0$$

$$\alpha = \frac{-3}{7}$$

$$98(\alpha + \beta)^2 = (98) \frac{(144)}{49} = 288$$